## MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 12

**1.** *Let* $\mathbf{v}_1 = \begin{bmatrix}1\\0\\1\\1\end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$ *and* $\mathbf{v}_3 = \begin{bmatrix}2\\1\\0\\-1\end{bmatrix}$. *Using Gram-Schmidt, find an orthonormal basis for* $W = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, *using* $\mathbf{v}_1, \mathbf{v}_2,$ *and* $\mathbf{v}_3$.

**2.** *Previously, we learned that given a basis of some subspace* $W$, *the Gram-Schmidt process produces an* orthonormal *basis of the same subspace* $W$. *In fact, we need not start with a basis—it is sufficient to start with any spanning set of* $W$, *even if the set is not linearly independent.*

*Let* $\mathbf{v}_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}2\\1\\1\end{bmatrix}$. *Apply Gram-Schmidt to* $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ *and obtain an orthonormal basis for the subspace* $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

**3.** *Compute the QR-factorization for each of the following matrices (you may assume the columns are linearly independent):*

(1) $\begin{bmatrix}1 & 2 & 5\\-1 & 1 & -4\\-1 & 4 & -3\\1 & -4 & 7\\1 & 2 & 1\end{bmatrix}$

(2) $\begin{bmatrix}1 & 1 & 1\\1 & 1 & 0\\1 & 0 & 0\\1 & 0 & 0\end{bmatrix}$

(3) $\begin{bmatrix}1 & 1 & 1\\0 & 1 & 1\\0 & 0 & 1\\0 & 0 & 1\end{bmatrix}$

**4.** *Let* $A = \begin{bmatrix}1 & 0\\0 & 1\\1 & 1\end{bmatrix}$.

**a.** *Find the QR decomposition of A: write* $A = QR$ *where* $Q$ *is a matrix with orthonormal columns and* $R$ *is an upper triangular matrix.*

**b.** *Let* $\mathbf{b} = \begin{bmatrix}0\\0\\1\end{bmatrix}$. *Use the QR decomposition of* $A$ *to find the least squares solution of* $A\hat{\mathbf{x}} = \mathbf{b}$ *(by solving* $R\hat{\mathbf{x}} = Q^T\mathbf{b}$*).*

**5.** *Let* $A = \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$.

(1) *Find a diagonal matrix $D$ and an invertible matrix $P$ such that $A = PDP^{-1}$.*
(2) *Can you choose $P$ to an be orthogonal matrix?*

**6.** *True/False. Give reasons why!*

(1) *The $QR$ decomposition of a matrix is unique.*
(2) *If $A = QR$, where $Q$ has orthonormal columns, then $R = Q^T A$.*
(3) *In a $QR$ factorization, say $A = QR$ (when $A$ has linearly independent columns), the columns of $Q$ form an orthonormal basis for the column space of $A$.*
(4) *If $A = QR$ (as in the $QR$ decomposition), then $Col(A) = Col\,(R)$.*